

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Blu-ray
Manufacturer: Denon
Firmware Version: Revision: 4797
Model(s): DN-500BD, DN-500BD MKII

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.13.0000-b006	2.13.0

Version History

Module Version	Date	Notes
1_3_2_0	8/25/2021	Fixed: - Status for overall driver Updated - Volume command range - Media Status value names Removed: - Connection Status - Serial pin diagram from Communication Sheet. FS Play state from Playback Status command.
1_2_1_0	4/25/2019	Added Subtitle command.
1_2_0_0	5/3/2018	Added Shutdown command and removed Power command for Ethernet control. Removed ConnectionStatus for Ethernet control.
1_1_0_0	2/7/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: InterfaceName.Unidirectional = 'True'
- For Ethernet control: Device cannot be controlled when the device is OFF because the Ethernet port closes when it is OFF.
- This module requires a 0.3 second command pacing for Serial and 0.03 second command pacing for Ethernet.

Supported Classes and Examples

SerialClass
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='DN-500BD')
SerialOverEthernetClass
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='DN-500BD')
EthernetClass
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9030, Model='DN-500BD')

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value		
ChapterorTrack	1 to 2000 in steps of 1		
# ChapterorTrack example InterfaceName.Set('ChapterorTrack', 2000)			
Command	Value	Value	Value
ColorFunction	'Red' 'Yellow'	'Green'	'Blue'
# ColorFunction example InterfaceName.Set('ColorFunction', 'Red')			
Command	Value		
CurrentTitle ¹	0 to 2000 in steps of 1		
# CurrentTitle example InterfaceName.Set('CurrentTitle', 2000)			
Command	Value	Value	
DiscTray	'Open'	'Close'	
# DiscTray example InterfaceName.Set('DiscTray', 'Open')			
Command	Value		
DisplayInfo	None		
# DisplayInfo example InterfaceName.Set('DisplayInfo', None)			
Command	Value	Value	
ExecutiveMode	'On'	'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command	Value		
Keypad	'0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command	Value	Value	Value
Language	'English' 'German' 'Italian' 'Swedish' 'Russian' 'Off'	'French' 'Dutch' 'Portuguese' 'Finnish' 'Korean'	'Spanish' 'Chinese' 'Danish' 'Norwegian' 'Japanese'
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Mode'	'Audio' 'Subtitle'	'Disc Menu'	'OSD'
# Language example InterfaceName.Set('Language', 'English', {'Mode': 'Audio'})			

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Command MenuNavigation	Value 'Setup Menu' 'Pop Up Menu' 'Down' 'Enter'	Value 'Disc Menu' 'Return' 'Left' 'Home'	Value 'Option Menu' 'Up' 'Right'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Setup Menu')			
Command Mute	Value 'On'	Value 'Off'	
# Mute example InterfaceName.Set('Mute', 'On')			
Command Power ²	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command PowerOff ⁴	Value None		
# PowerOff example InterfaceName.Set('PowerOff', None)			
Command RandomMode	Value 'Toggle'		
# RandomMode example InterfaceName.Set('RandomMode', 'Toggle')			
Command Search	Value None		
Qualifier Key 'Speed'	Qualifier Value 'Fast'	Qualifier Value 'Slow'	
Qualifier Key 'Direction'	Qualifier Value 'Forward'	Qualifier Value 'Reverse' ³	
# Search example InterfaceName.Set('Search', None, {'Speed': 'Fast', 'Direction': 'Forward'})			
Command Subtitle	Value None		
# Subtitle example InterfaceName.Set('Subtitle', None)			
Command Transport	Value 'Stop' 'Track/Chapter Next' 'Group/Title Prev'	Value 'Play' 'Track/Chapter Prev'	Value 'Pause' 'Group/Title Next'
# Transport example InterfaceName.Set('Transport', 'Stop')			
Command Volume	Value -90 to 0 in steps of 1 dB		
# Volume example InterfaceName.Set('Volume', 0)			

¹ Not all Media Type's support this command.

² Only for Serial control.

³ Reverse is only allowed when Speed is set to Fast.

⁴ Only for Ethernet control.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AlbumName ¹	Value 'String'
# AlbumName example InterfaceName.Update('AlbumName') Value = InterfaceName.ReadStatus('AlbumName') InterfaceName.SubscribeStatus('AlbumName', None, FeedbackHandler)	
Command ArtistName ¹	Value 'String'
# ArtistName example InterfaceName.Update('ArtistName') Value = InterfaceName.ReadStatus('ArtistName') InterfaceName.SubscribeStatus('ArtistName', None, FeedbackHandler)	
Command ChapterorTrack	Value 1 to 2000 in steps of 1
# ChapterorTrack example InterfaceName.Update('ChapterorTrack') Value = InterfaceName.ReadStatus('ChapterorTrack') InterfaceName.SubscribeStatus('ChapterorTrack', None, FeedbackHandler)	
Command ChapterorTrackTotal	Value 0 – 9999
# ChapterorTrackTotal example InterfaceName.Update('ChapterorTrackTotal') Value = InterfaceName.ReadStatus('ChapterorTrackTotal') InterfaceName.SubscribeStatus('ChapterorTrackTotal', None, FeedbackHandler)	
Command CurrentTitle ¹	Value 0 to 2000 in steps of 1
# CurrentTitle example InterfaceName.Update('CurrentTitle') Value = InterfaceName.ReadStatus('CurrentTitle') InterfaceName.SubscribeStatus('CurrentTitle', None, FeedbackHandler)	
Command CurrentTitleTotal ¹	Value 0 – 9999
# CurrentTitleTotal example InterfaceName.Update('CurrentTitleTotal') Value = InterfaceName.ReadStatus('CurrentTitleTotal') InterfaceName.SubscribeStatus('CurrentTitleTotal', None, FeedbackHandler)	

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Command	Value		
CurrentTrackTime	'String'		
# CurrentTrackTime example InterfaceName.Update('CurrentTrackTime') Value = InterfaceName.ReadStatus('CurrentTrackTime') InterfaceName.SubscribeStatus('CurrentTrackTime', None, FeedbackHandler)			
Command	Value	Value	Value
DeviceStatus	'Play'	'Pause'	'Slow Play Reverse'
	'Fast Play Reverse'	'Step Play'	'Menu'
	'Setup'	'Track Menu'	'Home'
	'Slow Play Forward'	'Fast Play Forward'	'Other'
# DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command	Value		
ElapsedTime	'String'		
# ElapsedTime example InterfaceName.Update('ElapsedTime') Value = InterfaceName.ReadStatus('ElapsedTime') InterfaceName.SubscribeStatus('ElapsedTime', None, FeedbackHandler)			
Command	Value	Value	Value
Language	'English'	'French'	'Spanish'
	'German'	'Dutch'	'Chinese'
	'Italian'	'Portuguese'	'Danish'
	'Swedish'	'Finnish'	'Norwegian'
	'Russian'	'Korean'	'Japanese'
	'Off'		
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Mode'	'Audio'	'Disc Menu'	'OSD'
	'Subtitle'		
# Language example InterfaceName.Update('Language', {'Mode': 'Audio'}) Value = InterfaceName.ReadStatus('Language', {'Mode': 'Audio'}) InterfaceName.SubscribeStatus('Language', None, FeedbackHandler)			
Command	Value	Value	Value
MediaStatus	'No Disc'	'Disc In'	'Unformatted Disc'
	'Tray Open'	'Tray Closed'	'Tray Error'
# MediaStatus example InterfaceName.Update('MediaStatus') Value = InterfaceName.ReadStatus('MediaStatus') InterfaceName.SubscribeStatus('MediaStatus', None, FeedbackHandler)			
Command	Value	Value	Value
MediaType	'DVD Video'	'DVD Audio'	'DVD VR'
	'CD-ROM'	'SACD'	'CDDA'
	'BDMV'	'BD-DAV'	'AVCHD'
	'DLNA'	'External Memory'	'Unknown'
# MediaType example InterfaceName.Update('MediaType') Value = InterfaceName.ReadStatus('MediaType') InterfaceName.SubscribeStatus('MediaType', None, FeedbackHandler)			

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Command Mute	Value 'On'	Value 'Off'
<pre># Mute example InterfaceName.Update('Mute') Value = InterfaceName.ReadStatus('Mute') InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)</pre>		
Command Power ³	Value 'On'	Value 'Off'
<pre># Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>		
Command RandomMode	Value 'Toggle' 'Off'	Value 'Shuffle' 'Random'
<pre># RandomMode example InterfaceName.Update('RandomMode') Value = InterfaceName.ReadStatus('RandomMode') InterfaceName.SubscribeStatus('RandomMode', None, FeedbackHandler)</pre>		
Command RemainingTime	Value 'String'	
<pre># RemainingTime example InterfaceName.Update('RemainingTime') Value = InterfaceName.ReadStatus('RemainingTime') InterfaceName.SubscribeStatus('RemainingTime', None, FeedbackHandler)</pre>		
Command Volume ²	Value -90 to 0 in steps of 1 dB	
<pre># Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>		

¹ Not all Media Type's support this command

² For the command to work, ensure the "Output Volume" setting is set to "Variable" in the device menu.
"Setup Menu" → "Audio Setup" → "Output Control" → "Output Volume" → "Variable"

³ Only available for Serial control

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

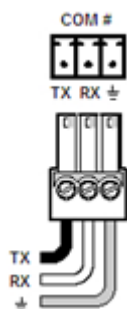
Data Bits: 8

Parity: None

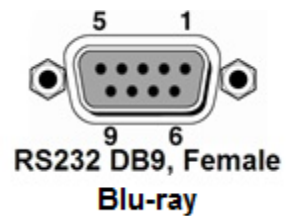
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Please contact the manufacturer for pin assignment diagram.



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 9030

Logon Credentials Supported: No

Multi-Connection Capabilities: No

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Chapter or Track 1	@0Tr0001\x0D
Chapter or Track 2000	@0Tr2000\x0D
Color Function Blue	@0DVFCLR3\x0D
Color Function Green	@0DVFCLR2\x0D
Color Function Red	@0DVFCLR1\x0D
Color Function Yellow	@0DVFCLR4\x0D
Current Title 0	@0PCGp0000\x0D
Current Title 2000	@0PCGp2000\x0D
Disc Tray Close	@0PCDTRYCL\x0D
Disc Tray Open	@0PCDTRYOP\x0D
Display Info None	@0DVDSIF\x0D
Executive Mode Off	@023KU\x0D
Executive Mode On	@023KL\x0D
Keypad 0	@0PCTKEY0\x0D
Keypad 9	@0PCTKEY9\x0D
Language Chinese Mode Audio	@0DVLGADzho\x0D
Language Chinese Mode Disc Menu	@0DVLGDMzho\x0D
Language Chinese Mode OSD	@0DVLGOSzho\x0D
Language Chinese Mode Subtitle	@0DVLGSTzho\x0D
Language Danish Mode Audio	@0DVLGADdan\x0D
Language Danish Mode Disc Menu	@0DVLGDMdan\x0D
Language Danish Mode OSD	@0DVLGOSdan\x0D
Language Danish Mode Subtitle	@0DVLGSTdan\x0D
Language Dutch Mode Audio	@0DVLGADnld\x0D
Language Dutch Mode Disc Menu	@0DVLGDMnld\x0D
Language Dutch Mode OSD	@0DVLGOSnld\x0D
Language Dutch Mode Subtitle	@0DVLGSTnld\x0D
Language English Mode Audio	@0DVLGADeng\x0D
Language English Mode Disc Menu	@0DVLGDMeng\x0D
Language English Mode OSD	@0DVLGOSeng\x0D
Language English Mode Subtitle	@0DVLGSTeng\x0D
Language Finnish Mode Audio	@0DVLGADfin\x0D
Language Finnish Mode Disc Menu	@0DVLGDMfin\x0D
Language Finnish Mode OSD	@0DVLGOSfin\x0D
Language Finnish Mode Subtitle	@0DVLGSTfin\x0D
Language French Mode Audio	@0DVLGADfra\x0D
Language French Mode Disc Menu	@0DVLGDMfra\x0D
Language French Mode OSD	@0DVLGOSfra\x0D
Language French Mode Subtitle	@0DVLGSTfra\x0D
Language German Mode Audio	@0DVLGADdeu\x0D
Language German Mode Disc Menu	@0DVLGDMdeu\x0D
Language German Mode OSD	@0DVLGOSdeu\x0D

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Language German Mode Subtitle	@0DVLGSTdeu\x0D
Language Italian Mode Audio	@0DVLGADita\x0D
Language Italian Mode Disc Menu	@0DVLGDMita\x0D
Language Italian Mode OSD	@0DVLGOSita\x0D
Language Italian Mode Subtitle	@0DVLGSTita\x0D
Language Japanese Mode Audio	@0DVLGADjpn\x0D
Language Japanese Mode Disc Menu	@0DVLGDMjpn\x0D
Language Japanese Mode OSD	@0DVLGOSjpn\x0D
Language Japanese Mode Subtitle	@0DVLGSTjpn\x0D
Language Korean Mode Audio	@0DVLGADkor\x0D
Language Korean Mode Disc Menu	@0DVLGDMkor\x0D
Language Korean Mode OSD	@0DVLGOSkor\x0D
Language Korean Mode Subtitle	@0DVLGSTkor\x0D
Language Norwegian Mode Audio	@0DVLGADnor\x0D
Language Norwegian Mode Disc Menu	@0DVLGDMnor\x0D
Language Norwegian Mode OSD	@0DVLGOSnor\x0D
Language Norwegian Mode Subtitle	@0DVLGSTnor\x0D
Language Off Mode Audio	@0DVLGADOFF\x0D
Language Off Mode Disc Menu	@0DVLGDMOFF\x0D
Language Off Mode OSD	@0DVLGOSOFF\x0D
Language Off Mode Subtitle	@0DVLGSTOFF\x0D
Language Portuguese Mode Audio	@0DVLGADpor\x0D
Language Portuguese Mode Disc Menu	@0DVLGDMpor\x0D
Language Portuguese Mode OSD	@0DVLGOSpor\x0D
Language Portuguese Mode Subtitle	@0DVLGSTpor\x0D
Language Russian Mode Audio	@0DVLGADrus\x0D
Language Russian Mode Disc Menu	@0DVLGDMrus\x0D
Language Russian Mode OSD	@0DVLGOSrus\x0D
Language Russian Mode Subtitle	@0DVLGSTrus\x0D
Language Spanish Mode Audio	@0DVLGADspa\x0D
Language Spanish Mode Disc Menu	@0DVLGDMspa\x0D
Language Spanish Mode OSD	@0DVLGOSspa\x0D
Language Spanish Mode Subtitle	@0DVLGSTspa\x0D
Language Swedish Mode Audio	@0DVLGADswe\x0D
Language Swedish Mode Disc Menu	@0DVLGDMswe\x0D
Language Swedish Mode OSD	@0DVLGOSswe\x0D
Language Swedish Mode Subtitle	@0DVLGSTswe\x0D
Menu Navigation Disc Menu	@0DVTP\x0D
Menu Navigation Down	@0PCCUSR4\x0D
Menu Navigation Enter	@0PCENTR\x0D
Menu Navigation Home	@0PCHM\x0D
Menu Navigation Left	@0PCCUSR1\x0D
Menu Navigation Option Menu	@0DVOP\x0D
Menu Navigation Pop Up Menu	@0DVPU\x0D

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Menu Navigation Return	@0PCRTN\x0D
Menu Navigation Right	@0PCCUSR2\x0D
Menu Navigation Setup Menu	@0PCSU\x0D
Menu Navigation Up	@0PCCUSR3\x0D
Mute Off	@0mt01\x0D
Mute On	@0mt00\x0D
Power Off	@0PW01\x0D
Power Off None	@0PW01\x0D
Power On	@0PW00\x0D
Random Mode Toggle	@0PCPMR\x0D
Search None Speed Fast Direction Forward	@0PCSLSFf\x0D
Search None Speed Fast Direction Reverse	@0PCSLSRf\x0D
Search None Speed Slow Direction Forward	@0PCSLSFs\x0D
Subtitle None	@0DVSBTL1\x0D
Transport Group/Title Next	@0PCGPNX\x0D
Transport Group/Title Prev	@0PCGPPV\x0D
Transport Pause	@02348\x0D
Transport Play	@02353\x0D
Transport Stop	@02354\x0D
Transport Track/Chapter Next	@02332\x0D
Transport Track/Chapter Prev	@02333\x0D
Volume 0	@0DVOV000\x0D
Volume -90	@0DVOV-90\x0D

Appendix B. Update Commands

Album Name	@0?a1\x0D
Artist Name	@0?at\x0D
Chapter or Track	@0?Tr\x0D
Chapter or Track Total	@0?Tt\x0D
Current Title	@0?PCGp\x0D
Current Title Total	@0?PCTG\x0D
Current Track Time	@0?t1\x0D
Device Status	@0?ST\x0D
Elapsed Time	@0?ET\x0D
Language Mode Audio	@0?DVLGAD\x0D
Language Mode Disc Menu	@0?DVLGDM\x0D
Language Mode OSD	@0?DVLGOS\x0D
Language Mode Subtitle	@0?DVLGST\x0D
Media Status	@0?CD\x0D
Media Type	@0?PCTYP\x0D
Mute	@0?mt\x0D
Power	@0?PW\x0D
Random Mode	@0?PCPMR\x0D
Remaining Time	@0?RM\x0D
Volume	@0?DVOV\x0D